

The empirical recurrence for OEIS A255157, recovered from its tail

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Abstract

OEIS A255157 records “Empirical recurrence of order 79” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 163$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 8$, so the recurrence is proved for every $n \geq 87$. Its last coefficient is $c_{79} = -1014 \neq 0$, so no further tail satisfies anything shorter; any order-79 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A255157 is “Number of $(n+2) \times (6+2)$ 0..1 arrays with no 3×3 subblock diagonal sum 1 and no antidiagonal sum 1 and no row sum 0 or 1 and no column sum 0 or 1.”. It has offset 1 and begins

1782, 9978, 67465, 437238, 2861823, 18908787, 124074113, 813938431, ...

The entry states:

Empirical recurrence of order 79 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 11:39:58, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 163$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 163$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 79: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 8$ is the first whose minimal order is exactly the stated 79.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 326$ exact terms from $n = 8$ on, it returns order 79 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{79} c_i a(n-i),$$

c_1	1	c_{21}	−227690208	c_{41}	−62816567269	c_{61}	−357252732
c_2	16	c_{22}	855014089	c_{42}	−48496359551	c_{62}	521791091
c_3	83	c_{23}	721021088	c_{43}	−33691135291	c_{63}	−502047614
c_4	371	c_{24}	−59206498	c_{44}	−55075192664	c_{64}	372490918
c_5	405	c_{25}	276549395	c_{45}	10696213379	c_{65}	−211547645
c_6	−1879	c_{26}	−3894878765	c_{46}	8751010865	c_{66}	99144691
c_7	−11118	c_{27}	−3870395403	c_{47}	−5927406079	c_{67}	−53392393
c_8	−26275	c_{28}	7336333946	c_{48}	64121112903	c_{68}	22957019
c_9	25478	c_{29}	4557260870	c_{49}	−37492478677	c_{69}	−4108455
c_{10}	229040	c_{30}	−3240630705	c_{50}	39370624183	c_{70}	752085
c_{11}	284602	c_{31}	4734236931	c_{51}	−22730719181	c_{71}	1238217
c_{12}	−405268	c_{32}	−12469519991	c_{52}	−6231139291	c_{72}	−1474638
c_{13}	−1550511	c_{33}	−26879555691	c_{53}	6388455212	c_{73}	930775
c_{14}	−1863226	c_{34}	11407114513	c_{54}	−18174221543	c_{74}	−278378
c_{15}	−2197091	c_{35}	−589363734	c_{55}	18677697489	c_{75}	−1886
c_{16}	2517291	c_{36}	9753278973	c_{56}	−13920962363	c_{76}	20843
c_{17}	31849405	c_{37}	53833513305	c_{57}	11997120028	c_{77}	−10478
c_{18}	62680415	c_{38}	−5010698152	c_{58}	−5802702481	c_{78}	2498
c_{19}	−42981709	c_{39}	1859871690	c_{59}	2679511367	c_{79}	−1014
c_{20}	−371294032	c_{40}	−11192970844	c_{60}	−802448363		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 87$, and it is the recurrence OEIS A255157 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 8$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{79} - c_1 x^{79-1} - \dots - c_{79}$. Its constant term is $-c_{79} = 1014$, which is not zero, so $q_{\min}(0) \neq 0$ and

$q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 79 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 79, so $\deg q = 79$, and it is monic. It holds from some index t on. From $\max(t, 8)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 79, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 18 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 79, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -1014 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-79 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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